🌐 Subnetting Part 2 & VLSM (10.28.2026)

Big networks, same sharp tools. This builds on Part 1 and focuses on Class B subnetting, prefix choices, and fast subnet/host math.

📐 Core Formulas

- **Subnets** = `2^b` *(where b = borrowed bits)*
- **Usable hosts** = `2^h - 2` *(where h = host bits left)*
- **Host bits** = `32 - prefix length`
- **Block size** = `256 - mask octet`

> 💡 **Memory Hook:** Borrow for subnets, leave for hosts. Find the block, land the host, write the network.

🧠 Quiz Answer Key + Quick Why

1. **Given 172.30.0.0/16, need 100 subnets with at least 500 hosts each. Prefix?**
   - **Answer:** `/23` or `255.255.254.0`
   - **Why:** Need 500 hosts ➡️ 9 host bits (`2^9 - 2 = 510`). `32 - 9 = 23` ➡️ /23
2. **Host 172.21.111.201/20 belongs to which subnet?**
   - **Answer:** `172.21.96.0`
   - **Why:** /20 mask = 255.255.240.0 ➡️ block size 16 in 3rd octet. `111` falls in the **96-111** block.
3. **Broadcast address of 192.168.91.78/26?**
   - **Answer:** `192.168.91.127`
   - **Why:** /26 mask = 255.255.255.192 ➡️ block size 64. `78` falls in the **64-127** block.
4. **Divide 172.16.0.0/16 into 4 equal subnets. Second subnet's network and broadcast?**
   - **Network:** `172.16.64.0`
   - **Broadcast:** `172.16.127.255`
   - **Why:** 4 subnets = borrow 2 bits ➡️ /18. Block size in 3rd octet = 64. Second subnet = **64-127**.
5. **Divide 172.30.0.0/16 into subnets of 1000 hosts each. How many subnets?**
   - **Answer:** `64 subnets`
   - **Why:** 1000 hosts ➡️ 10 host bits (`2^10 - 2 = 1022`). New prefix = /22 ➡️ borrowed bits = `22 - 16 = 6`. `2^6 = 64`.

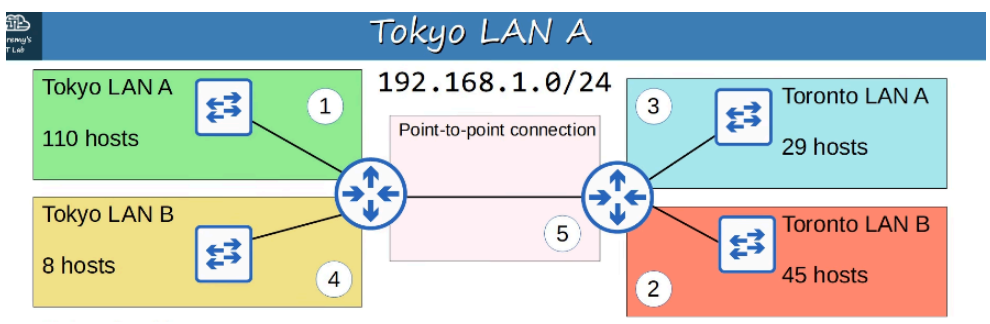
🧩 Variable-Length Subnet Masks (VLSM)

⚖️ FLSM vs. VLSM

- **FLSM (Fixed-Length):** All subnets use the *same* prefix length (e.g., dividing a network into four equal /26s). Wastes addresses!
- **VLSM (Variable-Length):** Creates subnets of *different* sizes to perfectly fit your needs. Highly efficient but requires careful planning.

🪜 The 3-Step VLSM Process

1. Assign the **largest** subnet at the start of the address space.
2. Assign the **second-largest** subnet immediately after it.
3. Repeat the process (largest to smallest) until all subnets are assigned.



🗺️ VLSM Practice Example: 192.168.1.0/24

Requirements: Tokyo A (110 hosts), Toronto B (45 hosts), Toronto A (29 hosts), Tokyo B (8 hosts), Point-to-Point (2 hosts).

Order / Subnet	Prefix	Network Address	Broadcast Address	Usable Range	Total Usable
1. Tokyo LAN A (110)	`/25`	`192.168.1.0`	`192.168.1.127`	`.1` - `.126`	126
2. Toronto LAN B (45)	`/26`	`192.168.1.128`	`192.168.1.191`	`.129` - `.190`	62
3. Toronto LAN A (29)	`/27`	`192.168.1.192`	`192.168.1.223`	`.193` - `.222`	30
4. Tokyo LAN B (8)	`/28`	`192.168.1.224`	`192.168.1.239`	`.225` - `.238`	14
5. Point-to-Point (2)	`/30`	`192.168.1.240`	`192.168.1.243`	`.241` - `.242`	2

> ⚠️ **Point-to-Point Note:** A `/31` prefix technically supports 2 hosts (RFC 3021) and saves space, but it is generally **not recommended** in standard curriculum/testing scenarios. Stick to `/30` for point-to-point links.